

Yuguang Li

yuguang.lee1@gmail.com | (315) 897-7889 | Seattle (Open to relocate) | [Website / Research Portfolio](#) | [Patent Portfolio](#)

Technical Profile: Principal Applied Scientist and PhD researcher specializing in multi-view perception, camera pose estimation, structured 3D reconstruction, and production ML systems. At Zillow, I led the technical development of the company's first production floor-plan reconstruction platform, scaling sparse wide-baseline indoor reconstruction from research prototypes into deployed workflows supporting hundreds of thousands of floor-plan-centric home experiences annually. My work spans pose estimation, room layout estimation, learned bundle adjustment, diffusion-conditioned optimization, image-based rendering, and production automation. Research output includes a first-author CVPR 2025 highlight paper, a first-author NeurIPS 2026 submission; several CVPR/ECCV publications with interns, large-scale datasets, deployed ML systems, and 30+ patents/patent filings.

Core Areas: Multi-view perception | 3D reconstruction | Camera pose estimation | Bundle adjustment / geometric optimization | Indoor scene understanding | Multimodal foundation models | Generative models / diffusion | Production ML systems | Graphics / image-based rendering | Geometric shape processing

EXPERIENCE

Zillow Group

Principal Applied Scientist / Technical Lead

Aug. 2017 – Present

Seattle

Continuously role since 2017: full-time 2017-2023 • Zillow-sponsored part-time PhD research 2024-2026 • returning full-time 2026

Focus: Production-scale indoor reconstruction from sparse, unposed, wide-baseline panoramas

Zillow Floor-plan Reconstruction Platform

 Originator • Technical Lead • System Architect (2018-2020)

- **Created demand for production-scale floor-plan media:** I originated the floor-plan-centered consumer experience, led the first-place Zillow Hack Week project, and patented the interaction concept that became the conceptual starting point for Zillow's [Showcase](#) experience. In parallel, I proved the underlying reconstruction workflow with a small annotator pilot, helping secure executive buy-in for broader investment in scalable floor-plan reconstruction; the resulting floor-plan media experience now supports hundreds of thousands of home experiences annually.
- **Originated the reconstruction platform:** I conceived Zillow's floor-plan reconstruction platform as a sparse multi-view indoor reconstruction system, defining the spatial representations, data contracts, reconstruction stages, and interactive geometry operations that still underlie the platform today. The system spans rotation correction, scale calibration, single-view layout estimation, camera-pose recovery/re-localization, multi-view geometry merging, and geometrically consistent floor-plan output. This platform later became the basis for the [ZInD dataset/system paper](#).
- **Built the precision reconstruction core:** Within this platform, I personally designed and prototyped the core CV/graphics system for localization, pose refinement, incremental room merging, and floor-plan geometry cleanup. This interactive reconstruction system made centimeter-level floor-plan production repeatable from sparse panoramas and became a key technical reason the workflow scaled to hundreds of thousands of homes per year before full automation.
- **Architected for modular ML automation:** I designed the platform around stable floor-plan representations and separable reconstruction stages, allowing learned components to be integrated as replaceable modules without changing the geometric output contract. This architecture let later ML models reduce high-volume verification/editing work while preserving centimeter-level accuracy at production scale.

Next-generation Autonomous Multi-view Reconstruction

 First-author Research Lead *BADGR* • *MvFFN/MOIS* (2024-2026)

- **Led a Zillow-backed autonomous reconstruction agenda:** Building on the production impact of the floor-plan platform I originated, I received leadership support to pursue autonomous floor-plan reconstruction at PhD depth, with broad technical freedom and collaboration across top-tier UW and senior industry researchers. I reframed the production pipeline as autonomous multi-view reconstruction from unposed, wide-baseline panoramas to centimeter-level multi-room floor plans with structured layouts.
- **First-authored [BADGR](#), CVPR 2025 Highlight:** *BADGR* targets centimeter-level floor-plan reconstruction from sparse wide-baseline images by combining reliable geometric refinement with learned priors for noisy or occluded regions. The method couples planar layout-pose bundle adjustment with a floor-plan specific *DDPM* diffusion model conditioned on Levenberg–Marquardt style optimization gradients, enabling learned outlier rejection and layout inpainting while preserving view-consistent geometry. I originated the core idea, implemented the method, and owned the experimental evaluation end to end.
- **First-authored [MvFFN/MOIS](#), NeurIPS 2026 submission:** I developed a first-of-its-kind feed-forward approach for autonomous floor-plan reconstruction from unposed, wide-baseline panoramas, targeting direct multi-view multi-room floor layout recovery without pose priors. After direct floor-plan prediction proved brittle under limited and noisy data, I reformulated the learning problem through our novel *MOIS* formulation and built *MvFFN* as a multi-

view geometry reasoning model. *MvFFN* combines geometric foundation-model features, semantic/geometric cross-view reasoning, dense segmentation/geometric losses, and two-view inductive biases to stabilize learning from masked, scarce, and inconsistently annotated data while recovering pixel-grounded cross-view wall identities, ordered wall structure, and within-room connectivity for downstream floor-plan assembly.

- **Built the data foundation for autonomous reconstruction:** I created the **ZInD-CrossView** dataset with panorama images, globally unique wall instances, global layouts, and dense cross-view segmentation labels using custom annotation tooling, enabling the *MvFFN/MOIS* work. I also assembled roughly 60k semantic floor plans as the **FloorPlan-60K** dataset to train *BADGR* diffusion model, using complementary data sources for different parts of the system: dense cross-view labels for multi-view instance reasoning and large semantic floor-plan corpora for generative layout priors.
- **Driving productization back into Zillow:** After the research phase, I am now returning this autonomous reconstruction stack to production as a full-time technical leader, redefining the floor-plan automation strategy with Zillow's floor-plan organization. I am translating *BADGR / MvFFN* outputs into production-compatible geometry, validation metrics, failure-mode analysis, and staged integration with the existing floor-plan automation pipeline.

Production ML Automation for Human-in-the-loop Reconstruction (2021-2023)

- **Owned the floor-plan ML automation agenda:** As the principal scientist driving floor-plan automation, I defined the automation roadmap across pose, layout, relocalization, and geometry cleanup; estimated the cost-saving potential of each module; led recurring reviews with PMs, scientists, and engineers; and tracked progress using metrics from UI-action counts to end-to-end annotation time. The combined automation program reduced floor-plan production from roughly 50 minutes in the manual proof-of-concept to roughly 15 minutes in the ML-assisted workflow.
- **Defined and mentored pose automation research:** I identified wide-baseline indoor pose estimation as a core automation problem and initiated research efforts for featureless scenes, extreme viewpoint changes, and sparse panorama overlap. As second author and senior technical mentor on both ECCV 2022 papers, I shaped the directions behind *SALVe*, a birds'-eye view (BEV) / semantic pose-estimation approach with pose-graph optimization, and *CovisPose*, a feed-forward co-visibility pose estimator for sparse panoramas that anticipated later correspondence-style pose methods such as *DUST3R*, in a 2D panorama setting.
- **Productized *CovisPose* end-to-end:** I drove *CovisPose* from research model to production localization system, including model scaling, training-data expansion, large-scale data re-annotation and cleanup, engineering integration, UI/design iteration, operator-facing prediction presentation, metric tracking, and user-feedback loops. The deployed system improved primary panorama selection and coarse floor-plan regeneration, reducing floor-plan annotation time by roughly 30%.
- **Originated *PSM-Net* for multi-view layout reconstruction:** I conceived the BEV-based multi-view layout estimation approach behind *PSM-Net*, built the initial proof of concept into a trainable model, and then recruited and mentored an intern and junior scientist to mature the method into a second-author CVPR 2022 publication. The work translated production floor-plan reconstruction needs into a publishable formulation for multi-view structured layout estimation.
- **Improved single-view layout automation:** I co-led the production iteration of the monocular room-layout model with a junior scientist, improving the model through stronger backbones, loss design, uncertainty estimation, and data hygiene. These changes made layout predictions more reliable inside the verification/editing workflow and contributed to roughly 25% reduction in layout-related annotation time.

Image-based Rendering and Virtual Touring Technical Lead • Graphics Prototype

- **Built low-cost virtual touring graphics:** designed an OpenGL-shader-based rendering approach, later patented with me as first-named inventor, for immersive indoor transitions between sparse, wide-baseline panoramas using floor-plan geometry and predicted depth instead of depth sensors or dense high-fidelity 3D reconstruction.
- **Connected reconstruction to consumer media:** I partnered with front-end and full-stack engineers to integrate floor-plan geometry, camera poses, panorama imagery, and shader-based transition effects into floor-plan-centric virtual touring experiences.

Leadership / IP: Principal scientist and technical lead across floor-plan reconstruction, pose estimation, layout estimation, autonomous reconstruction, and visual AI; senior mentor on CVPR/ECCV publications with interns and junior scientists; inventor on 30+ patents/patent filings covering floor-plan reconstruction, virtual touring, camera pose estimation, image-based rendering, and AI-assisted media.

University of Washington – Zillow-sponsored PhD Research Collaboration

Full-time PhD Researcher / Computer Vision & AI

Jan. 2024 – Jun. 2026

Seattle

- **Zillow-sponsored research track:** after originating and scaling Zillow's floor-plan reconstruction platform, I received Zillow leadership support to pursue the full autonomous reconstruction problem at academic depth while remaining tied to Zillow's long-term production roadmap. PhD defense scheduled for June 2026.

- **Cross-industry and UW collaboration:** collaborated with UW researchers, PhD students, and senior industry researchers across Zillow, Apple, NVIDIA, and Microsoft on *BADGR* and *MvFFN/MOIS*; contributed research direction, mentoring, system formulation, and technical integration. Detailed first-author contributions are listed under the Zillow autonomous reconstruction section above.

Adobe Creative Technology Labs

Researcher

June 2016 – Sept. 2016

Seattle

- Prototyped a high-performance cross-platform photo processing pipeline based on Halide compiler. Our work was later adopted as the computational photography framework in products like Photoshop, Lightroom, etc.

EDUCATION

PhD, Computer Vision/Artificial Intelligence, University of Washington

2024 - June 2026

MS, Computer Vision, University of Washington

MS, Geospatial Information Science, State University of New York

BE, Opto-electrical engineering, Beijing University of Aeronautics and Astronautics (top-5 in China)

TECHNICAL SKILLS

ML / Vision: PyTorch, model training/evaluation, diffusion models, multimodal foundation models, segmentation, pose estimation, bundle adjustment, geometric optimization.

Systems / Productization: Python, C++, AWS-based infrastructure, production ML deployment, data/annotation pipelines, experiment tracking, metric design.

Graphics / UI: OpenGL/GLSL shaders, image-based rendering, React, front-end prototyping.